Phy 211

Lectures Notes

By

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Lecture 1

Fall 2024

Chapter 1

Electric Force & Electric Field

<u>Electricity:</u> The art of (branch of physics) dealing with charges

<u>Electrostatic</u>

- Branch of physics dealing with charges at rest

<u>Dynamic electricity</u>

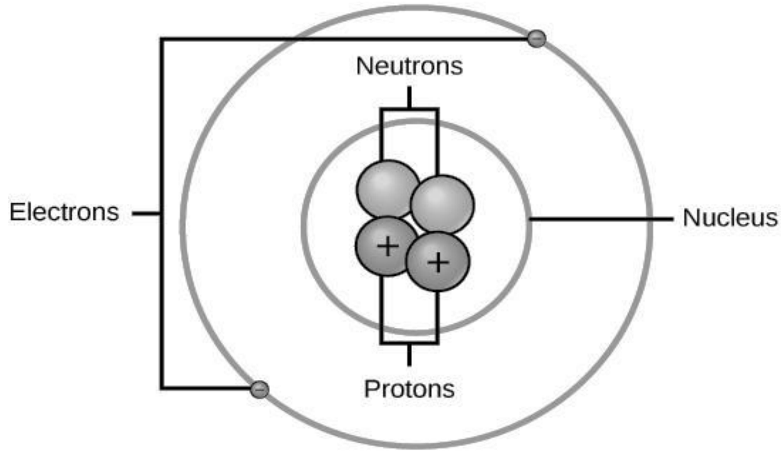
- Branch of physics dealing with charges at motion (electric current)
- Electric current produces magnetic field

- In 1909 Robert Millikan (1886–1953) discovered that if an object is charged, its charge is always a multiple of a **fundamental unit of charge**, given the symbol ***e*** *"electron charge"* .

$$e = 1.6 \times 10^{-19}\ \mathrm{C}$$

- The symbol of charge is q or Q

➢ **Properties of electric charge:**

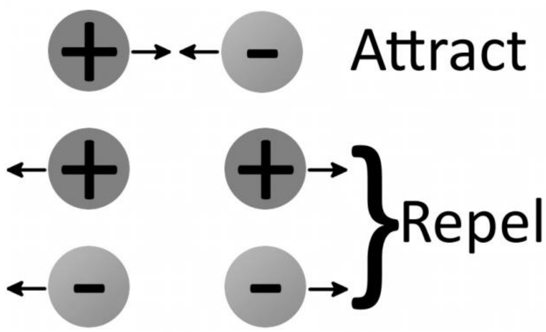


1- There are two types (kinds) of charge: positive "+ve" & negative "-ve" charge

2- Electron "e^-" is the smallest -ve charge ($q_e = -e = -1.6 \times 10^{-19}$ C)

3- Proton "p" is the smallest +ve charge $\left(q_p = +e = +1.6 \times 10^{-19}\ \mathrm{C}\right)$

4- Two similar charges (of same sign) repel each other, and two different charges (of opposite sign) attract each other



<u>Repulsive Force:</u> force between similar charges

<u>Attractive Force:</u> force between opposite charges

5- The SI unit of electric charge is the Coulomb (C).

6- Charge is quantized: charge consists of multiple integers of electron charge

$$\boxed{\boldsymbol{Q = \pm Ne}} \qquad \boldsymbol{Q = \pm 1e, \pm 2e, \pm 3e, \ldots\ldots}$$

$$\boldsymbol{Q: total\ charge} \qquad \boldsymbol{N: no.\ of\ e_s^- or\ p_s} \qquad \boldsymbol{e: fundamental\ charge}$$

- **Example 1:** A charged sphere with charge $(-64\ \mu\mathrm{C})$, calculate the no. of electrons on that sphere?

Solution

$$Q = -64\ \mu\mathrm{C} = -64 \times 10^{-6}\ \mathrm{C}$$

$$Q = -Ne \longrightarrow N = \frac{Q}{-e} = \frac{-64 \times 10^{-6}}{-1.6 \times 10^{-19}} = 40 \times 10^{13}\ electrons$$

- **Example 2:**

How many electrons are there in $-1\ \mathrm{mC}, -1\ \mu\mathrm{C}, -1\ \mathrm{nC}\ and - 1\ \mathrm{pC}$?

Solution

For $-1\ \mathrm{mC}$: $$Q = -1\ \mathrm{mC} = -1 \times 10^{-3}\ \mathrm{C}$$

$$N = \frac{Q}{e} = \frac{-1 \times 10^{-3}}{-1.6 \times 10^{-19}} = 6.25 \times 10^{15}\ electrons$$

For $-1\ \mu\mathrm{C}$: $$Q = -1\ \mu\mathrm{C} = -1 \times 10^{-6}\ \mathrm{C}$$

$$N = \frac{Q}{e} = \frac{-1 \times 10^{-6}}{-1.6 \times 10^{-19}} = 6.25 \times 10^{12}\ electrons$$

For $-1\ \mathrm{nC}$: $$Q = -1\ \mathrm{nC} = -1 \times 10^{-9}\ \mathrm{C}$$

$$N = \frac{Q}{e} = \frac{-1 \times 10^{-9}}{-1.6 \times 10^{-19}} = 6.25 \times 10^{9}\ electrons$$

For $-1\ \mathrm{pC}$: $$Q = -1\ \mathrm{pC} = -1 \times 10^{-12}\ \mathrm{C}$$

$$N = \frac{Q}{e} = \frac{-1 \times 10^{-12}}{-1.6 \times 10^{-19}} = 6.25 \times 10^{6}\ electrons$$

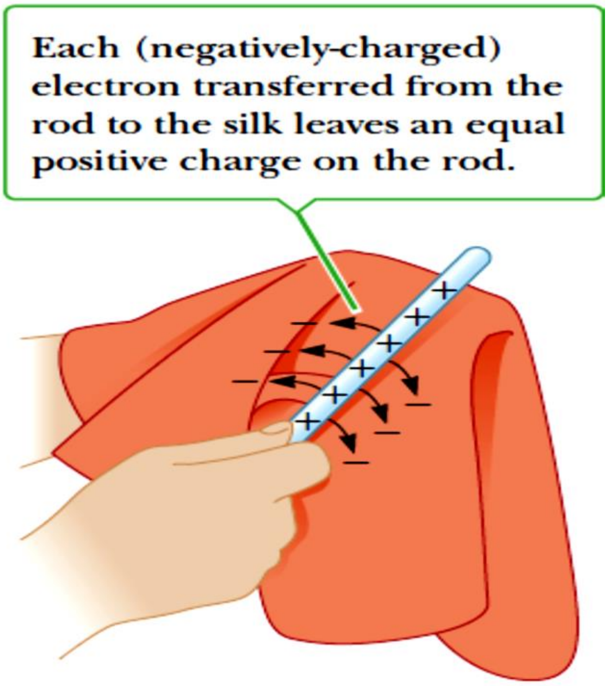
Unit	Symbol	conversion
1 milli C	1 mC	$1\ mC = 10^{-3}\ C$
1 micro C	1 μC	$1\ \mu C = 10^{-6}\ C$
1 nano C	1 nC	$1\ nC = 10^{-9}\ C$
1 pico C	1 pC	$1\ pC = 10^{-12}\ C$

➢ Charging methods:

Rubbing **Induction** **Conduction**

I- Rubbing (Friction):

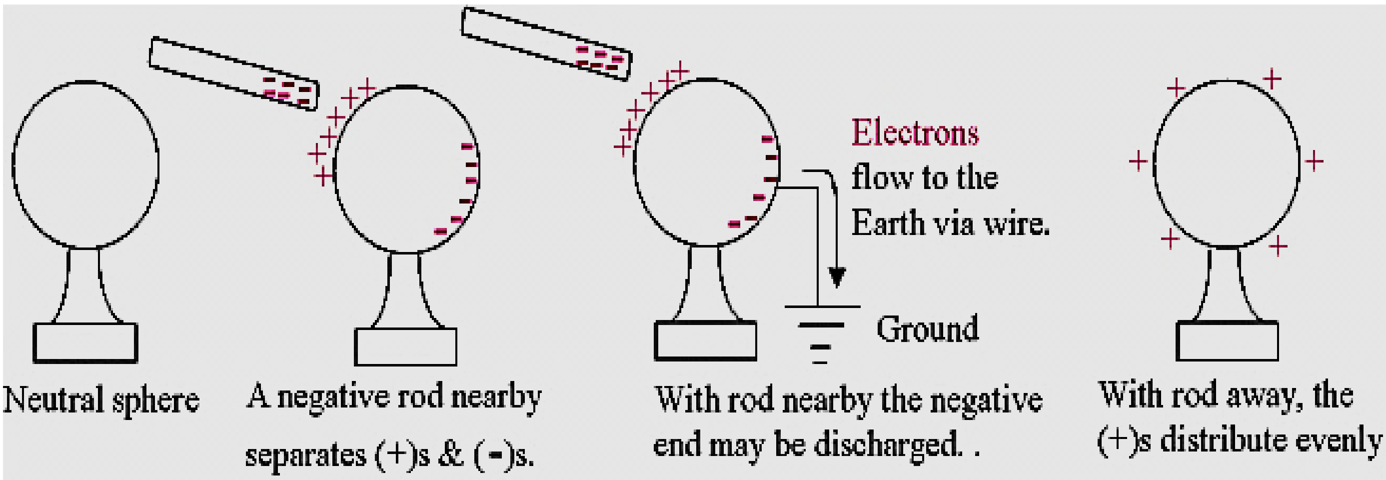
- When a glass rod is rubbed with silk, as in Figure, electrons are transferred from the rod to the silk. As a result, the glass rod carries a net +ve charge, the silk a net -ve charge.



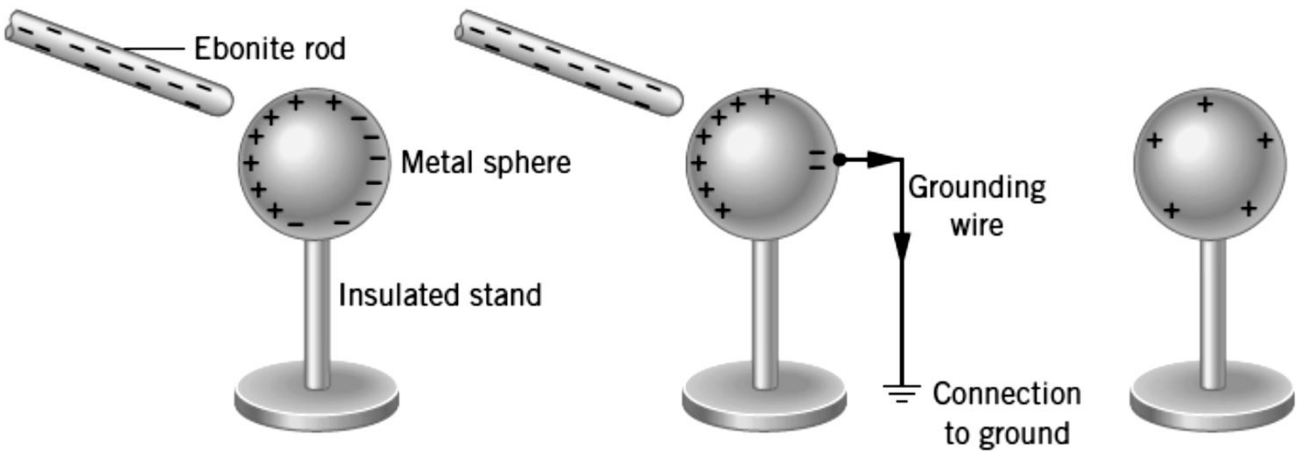
- Likewise, when plastic rod is rubbed with wool, electrons are transferred from the wool to the plastic. As a result, the plastic rod carries a net -ve charge, the wool a net +ve charge.
- This method is used for charging insulators.

II- Induction (for charging conductors):

- No direct contact is required between charging and charged bodies.
- If a plastic rod is rubbed against wool, it becomes negatively charged.

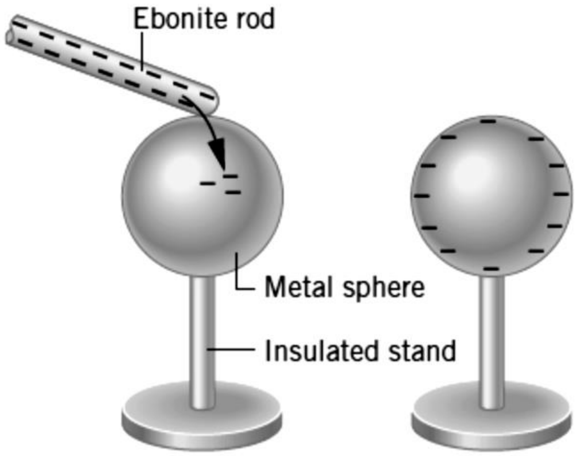


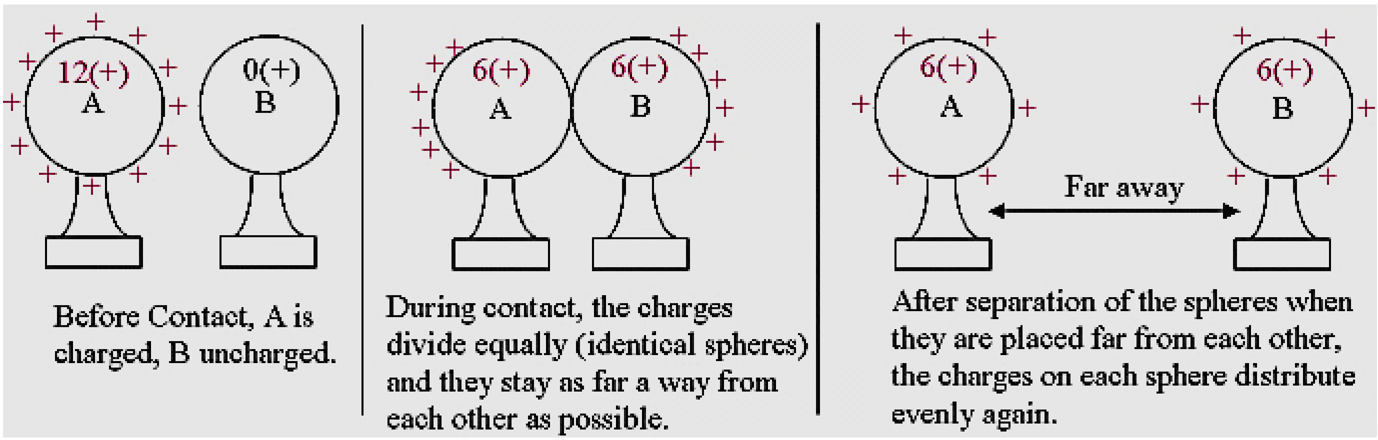
- If the rod is brought near a neutral metal sphere that is on an insulator mounting, it repels the free electrons of the sphere to the far end of it. At the same time, the rod attracts the positive charges of the sphere to the very near end of it.
- If the far end is connected to the Earth by a wire, the electrons flow to the ground while the positive charges are held by the rod.
- When the connection with the ground is cut off, the rod may be taken away leaving the sphere with positive charges.
- To charge an object positively by induction we use –ve charged object
- To charge an object negatively by induction we use +ve charged object



III- Conduction (for charging conductors):

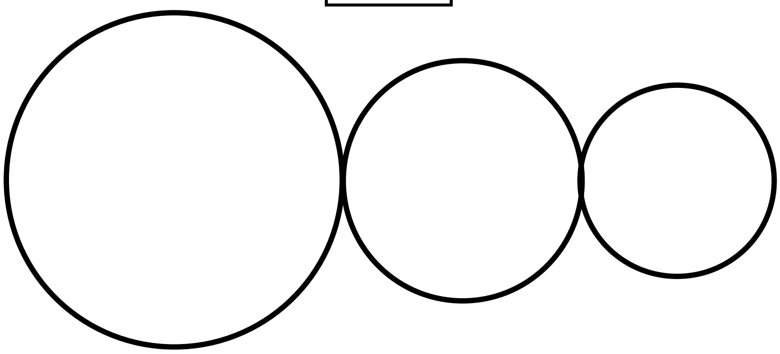
- Direct contact is required between charging and charged bodies.
- When a charged object is brought into contact with an uncharged (electrically neutral) object, part of its charges flow onto the uncharged object and make it partially charged.





- To charge an object positively by conduction we use +ve charged object
- To charge an object negatively by conduction we use -ve charged object
- The transfer proportion depends on the shapes of the two objects. For example, if the two objects are two identical metal spheres with insulator mountings, they share the charge equally.

$$\boxed{Q \propto r}$$



- After contact, the charge per unit radius "q'" is:

$$\boxed{\boldsymbol{q' = \frac{q_t}{r_t} = \frac{q_1 + q_2 + q_3 + \cdots}{r_1 + r_2 + r_3 + \cdots}}}$$

- The charge on the ith sphere is:

$$\boxed{\boldsymbol{q_i{}' = q' \times r_i = \frac{q_t}{r_t} \times r_i \quad \& \, i = 1, 2, 3, \ldots}}$$

$$\boxed{\boldsymbol{\frac{q_1{}'}{q_2{}'} = \frac{r_1}{r_2} \qquad \frac{q_1{}'}{q_3{}'} = \frac{r_1}{r_3} \qquad \frac{q_2{}'}{q_3{}'} = \frac{r_2}{r_3}}}$$

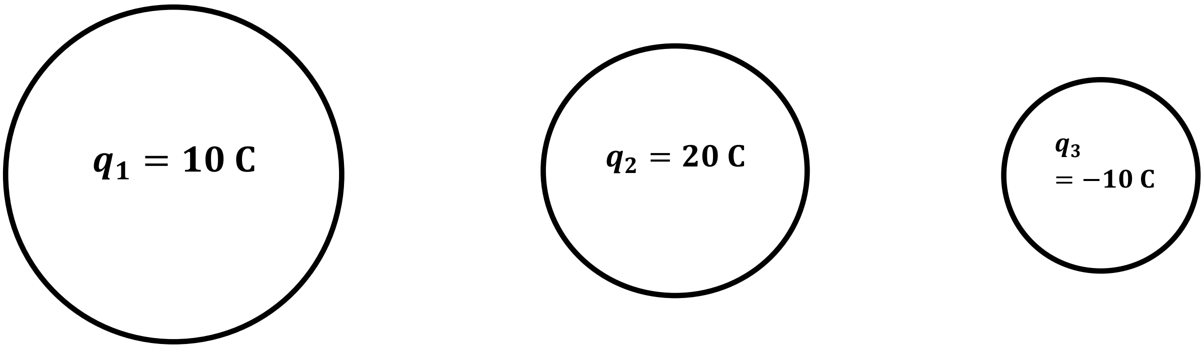
$$\boldsymbol{r_1, r_2, r_3 : radii\ of\ spheres\ 1, 2\ \&\ 3\ respectively}$$

$$\boldsymbol{q_1, q_2, q_3 : charge\ of\ spheres\ 1, 2\ \&\ 3\ respectively\ before\ contact}$$

$$\boldsymbol{q_1{}', q_2{}', q_3{}' : charge\ of\ spheres\ 1, 2\ \&\ 3\ respectively\ after\ contact}$$

- **Example 3:** Charged spheres having radii $3\ \mathrm{m}, 2\ \mathrm{m}\ \&\ 1\ \mathrm{m}$ and with different charges as shown in figure below. If we touch the three spheres to each other, find the final charge of each sphere.

Solution



$$q' = \frac{q_t}{r_t} = \frac{q_1 + q_2 + q_3}{r_1 + r_2 + r_3} = \frac{10 + 20 - 10q}{3 + 2 + 1} = \frac{20}{6}\ \mathrm{C/m}$$

$$q_1{}' = q' \times r_1 = \frac{20}{6} \times 3 = 10\ \mathrm{C}$$

$$q_2{}' = q' \times r_2 = \frac{20}{6} \times 2 = 6.667\ \mathrm{C}$$

$$q_3{}' = q' \times r_3 = \frac{20}{6} \times 1 = 3.333\ \mathrm{C}$$
